

K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) Classification

Problem Statement

You are provided with the famous [Iris flower dataset](#), which contains measurements of different iris flowers and their species.

Your task is to:

1. Preparing the dataset.
2. Implementing KNN classification from scratch (**no sklearn, no pandas**).
3. Tuning the parameter k and comparing validation accuracy.
4. Testing the best model on the test set.
5. Reporting all results.

About the Dataset: Iris Dataset

Property	Details
Number of Instances	150 samples (50 from each class)
Number of Attributes	4 numeric features + 1 class label
Features	Sepal length (cm), Sepal width (cm), Petal length (cm), Petal width (cm)
Target Classes	Iris-setosa, Iris-versicolor, Iris-virginica

Step 1: Load the dataset into a 2D Python list.

Step 2: Randomly split the dataset into:

- Training set: 70%
- Validation set: 15%
- Test set: 15%

Use random numbers to assign samples to these groups.

```
Train_set = [], Val_set = [], Test_set = []
```

For each sample S in D:

 Generate random number R in [0, 1]

 If $R \geq 0$ and $R \leq 0.7$:

 Add S to Train_set

 Else if $R > 0.7$ and $R \leq 0.85$:

 Add S to Val_set

 Else:

 Add S to Test_set

KNN Classification Tasks

For each sample V in Validation set:
For each sample T in Training set:
 Compute Euclidean distance between V's features and T's features
 Add (T, distance) to list L
Sort list L in ascending order by distance
Take the first k samples from L (nearest neighbours)
Count the majority class among these k samples → predicted_class
Now, check if this class is correct or not

Validation Accuracy = (correct_predictions / total validation samples) * 100

- [Code Template](#)
- Use k = 5 initially.
- For each validation sample:
 1. Compute Euclidean distance to all training samples.
 2. Sort training samples by distance.
 3. Select the top k neighbours.
 4. Assign the majority class among neighbours as the predicted class.
 5. Check if the prediction matches the true label.
- Calculate validation accuracy as:

$$\frac{\text{correct validation samples}}{\text{Total validation samples}} * 100$$

- Repeat the process for: k = 1, 3, 5, 10, 15
Make a results table:

K Value	Validation Accuracy (%)
1	
3	
5	
10	
15	

- Choose the best k (highest validation accuracy).

- Apply this best k on the test set to report final test accuracy.
- **Do NOT use: Libraries like sklearn, scikit-learn, or pandas for the algorithm, only for loading the raw data if needed.**
- Ensure your code is well-commented and your report is clear.
- **Any copying or plagiarism will result in -100% penalty.**

Marks Distribution

Task	Marks
Dataset loading	4
Train/Validation/Test split	5
KNN classification algorithm implementation	10
K tuning + validation accuracy table	3
Best k test accuracy calculation	3
Total	25